RetailEL: A Hybrid Lexical–Reranking Pipeline for Real-Time Grocery POS Entity Linking

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***Abstract*—Point-of-sale (POS) terminals in grocery retail produce heavily truncated, uppercased, and abbreviated product descriptions that must be linked to canonical stock-keeping unit (SKU) records for inventory analytics, demand forecasting, and receipt digitisation. Existing entity linking (EL) systems, whether neural bi-encoder models or large language model (LLM) pipelines, impose latency and computational costs incompatible with real-time transaction throughput. We present RetailEL, a five-stage hybrid lexical-reranking pipeline that achieves 97.82% Accuracy@1 on SynEL-500, a synthetic benchmark of 500 grocery SKUs with simulated POS noise, while maintaining a mean wall-clock latency of 4.28 ms per query (233.6 items/s). The pipeline comprises: (1) text normalisation via a 47-entry grocery abbreviation lexicon, (2) basket-context enrichment inspired by row-column entity linking, (3) hybrid BM25 + TF-IDF sparse retrieval fused with Reciprocal Rank Fusion, (4) a LightGBM LambdaRank reranker over eight lexical features, and (5) a finite-state-machine (FSM) constrained selection layer that provides a hard zero-hallucination guarantee. RetailEL surpasses a neural bi-encoder baseline (MiniLM-L6-v2, 84.51%) by 13.31 percentage points and bypasses LLM calls for 98.73% of queries, demonstrating that carefully engineered sparse retrieval with learned reranking can match or exceed neural approaches in domain-specific entity linking at a fraction of the cost.**

***Index Terms*—Entity Linking, Point-of-Sale, Grocery Retail, BM25, LambdaRank, Reranking, Sparse Retrieval, LightGBM**

I. Introduction

Entity linking in grocery retail is a surprisingly tricky problem that has real consequences for how stores manage stock, forecast demand, and process receipts. At the point-of-sale terminal, product descriptions come out short, all-caps, and packed with abbreviations—something like `ORG VLY GRK YGRT VNLA 32OZ` is supposed to stand in for "Organic Valley Greek Yogurt Vanilla 32oz." The gap between what the terminal prints and what the product database actually contains is wide enough that simple string matching does not cut it [8].

The core difficulty is that legacy POS hardware enforces hard character limits, typically 30–50 characters, which forces aggressive truncation. Domain-specific shorthands like `CHKN` for chicken or `BRST` for breast drop in naturally, but they are not standardised across stores or terminals. Function words and brand qualifiers disappear entirely. The result is a surface form that looks almost nothing like the canonical name it is meant to represent.

There are a few broad ways people have tried to tackle this. Neural bi-encoder models [6] embed mentions and candidates in a shared vector space and retrieve by similarity, which works reasonably well in general settings but needs GPU inference and tends to run 20–50 ms per query. That is too slow when a checkout system is pushing hundreds of items a second. LLM-based approaches [2], [3] can handle tricky disambiguation with in-context examples, but latencies above 500 ms and the risk of generating entity names that do not exist in the database are hard to justify in production. Sparse methods like BM25 are fast but lack the discriminative power to tell apart candidates that differ only in subtle ways after abbreviation.

We built **RetailEL** to sit in the space these approaches leave open: fast enough for real-time POS use, accurate enough to be useful, and guaranteed never to output an SKU that is not in the knowledge base. Our contributions are:

- A **five-stage pipeline** combining text normalisation, basket-context enrichment, hybrid sparse retrieval with Reciprocal Rank Fusion (RRF), LightGBM LambdaRank reranking, and FSM-constrained selection.
- **97.82% Accuracy@1** on SynEL-500, beating a neural bi-encoder baseline by 13.31 percentage points at 4.28 ms mean latency.
- A **zero-hallucination guarantee**: every prediction is constrained to a valid SKU, which matters a lot when a wrong match cascades into billing or inventory errors.
- **Ablation studies** that honestly quantify what each stage and each feature contributes, including cases where a component does not help.

II. Related Work

A. Entity Linking in Structured and Tabular Data

Most entity linking research targets unstructured text, but structured settings have attracted growing attention. Wang et al. [1] introduced RoCEL, a framework that links entities in table cells by encoding both row and column context together. The key insight is that the other entities sharing a row or a table are informative disambiguation signals—if one cell says "milk" and another says "cereal," you have a reasonable guess about what category the remaining cells belong to.

We borrowed that intuition for our basket-context enrichment stage: the other items in a shopping transaction play the same role that co-occurring cells play in a table.

B. LLM-Based Entity Linking

Liu et al. [2] proposed OneNet, which uses an LLM-based consensus mechanism that aggregates judgments across multiple candidate assessments to pick the correct entity, achieving strong performance with minimal labelled examples. Lin et al. [3] built LLM4BioEL for the biomedical domain, adding constrained decoding so the model cannot generate entity names outside the knowledge base, and an entropy-based contrastive decoding step to improve disambiguation. Both systems demonstrate that LLMs handle ambiguous cases well, but the inference costs make them impractical as a primary mechanism for real-time POS processing.

C. Learning-to-Rank

The reranking component of RetailEL builds on pairwise learning-to-rank methods. Burges et al. [5] introduced RankNet, which learns pairwise preferences from labelled data using a cross-entropy loss and gradient descent. LambdaRank [4] extended this by defining gradient updates that account for how swapping a pair of documents changes the target ranking metric, effectively letting the model optimise NDCG directly. We use LambdaRank through LightGBM's gradient boosting implementation.

D. Sentence Embeddings

Reimers and Gurevych [6] showed that fine-tuning BERT with siamese and triplet network objectives produces sentence embeddings that are much more useful for semantic similarity search than the raw BERT outputs. We use the MiniLM-L6-v2 checkpoint as a neural baseline to give a sense of how a standard dense retrieval approach handles the POS domain.

III. Methodology

The RetailEL pipeline processes each POS mention m through five sequential stages to produce a linked SKU $\hat{e}$ from a knowledge base $\mathcal{K} = \{e_1, e_2, \ldots, e_N\}$ of $N = 500$ canonical product names.

A. Stage 1: Text Normalisation

The first stage expands abbreviations using a manually curated 47-entry lexicon $\mathcal{L} = \{(a_i, x_i)\}_{i=1}^{47}$ built from common grocery POS patterns—entries like `ORG` $\rightarrow$ `ORGANIC`, `GRK` $\rightarrow$ `GREEK`, and `OZ` $\rightarrow$ `OUNCE`. We apply case folding, whitespace normalisation, and greedy left-to-right expansion to produce the normalised mention $m' = \textsc{Normalise}(m, \mathcal{L})$. This stage runs in a mean of 0.092 ms.

B. Stage 2: Basket Context Enrichment

Drawing from the row–column encoding idea in RoCEL [1], we concatenate the normalised descriptions of up to four co-purchased items in the same transaction to form an enriched mention:

$$m'' = m' \oplus \textsc{Sep} \oplus c_1 \oplus \textsc{Sep} \oplus \cdots \oplus c_k, \quad k \leq 4 \tag{1}$$

The idea is that what someone buys together—bread and butter, for instance—can help resolve category-level ambiguity when the mention itself is too short to be distinctive. In practice, this component's effectiveness depends heavily on how meaningful the co-purchase data is, a point we return to in Section V.

C. Stage 3: Hybrid Sparse Retrieval with RRF

We retrieve the top-k candidates from two sparse models, BM25 and TF-IDF, and merge the result lists using Reciprocal Rank Fusion (RRF):

$$\text{RRF}(e) = \sum_{r \in \mathcal{R}} \frac{1}{\kappa + \text{rank}_r(e)} \tag{2}$$

where $\mathcal{R} = \{\text{BM25}, \text{TF-IDF}\}$ and $\kappa = 60$. The top-20 candidates by fused score are forwarded to the reranker. Retrieval accounts for the bulk of the pipeline's latency at 3.529 ms.

D. Stage 4: LightGBM LambdaRank Reranker

The reranker is a LightGBM model trained with the LambdaRank objective [4], which directly optimises NDCG by computing gradient updates that reflect how swapping candidate pairs changes the ranking metric. For each of the 20 candidates e_j paired with the normalised mention m', we compute eight features:

1) **Levenshtein Distance**: $\text{LD}(m', e_j)$
2) **Levenshtein Ratio**: $1 - \frac{\text{LD}(m', e_j)}{\max(|m'|, |e_j|)}$
3) **Jaccard Token Overlap**: $\frac{|T_{m'} \cap T_{e_j}|}{|T_{m'} \cup T_{e_j}|}$ over word tokens
4) **Bigram Overlap**: $\frac{|B_{m'} \cap B_{e_j}|}{|B_{m'} \cup B_{e_j}|}$ over character bigrams
5) **Char-level Jaccard**: $\frac{|C_{m'} \cap C_{e_j}|}{|C_{m'} \cup C_{e_j}|}$ over individual characters
6) **Length Ratio**: $\min\left(\frac{|m'|}{|e_j|}, \frac{|e_j|}{|m'|}\right)$
7) **RRF Score**: The fused retrieval score from Stage 3
8) **Prior SKU Frequency**: Empirical frequency of e_j in the training set

The reranker selects $\hat{e} = \arg\max_{e_j} f_\theta(\mathbf{x}_j)$ in a mean of 0.590 ms.

E. Stage 5: FSM-Constrained Selection

Before returning a prediction, a finite-state machine verifies that $\hat{e} \in \mathcal{K}$. This provides a hard zero-hallucination guarantee: RetailEL cannot produce an entity name that does not exist in the knowledge base, unlike LLM-based systems where the decoder can generate plausible-looking but entirely fictional product names. The lookup takes a mean of 0.003 ms.

IV. Experimental Setup

A. Dataset: SynEL-500

We constructed SynEL-500 from the Instacart Online Grocery Shopping Dataset 2017 [7], sampling 500 canonical product names across 13 categories: Bakery, Beverages, Breakfast, Canned, Dairy, Deli, Frozen, Household, Meat, Pantry,

TABLE I
ACCURACY@1 ON SYNEL-500 TEST SET

System	Acc@1 (%)	Δ vs RetailEL
BM25-Only	74.53	−23.29
RRF Top-1 (No Reranker)	74.89	−22.93
Neural Bi-Encoder (MiniLM)	84.51	−13.31
RetailEL (Full Pipeline)	**97.82**	—

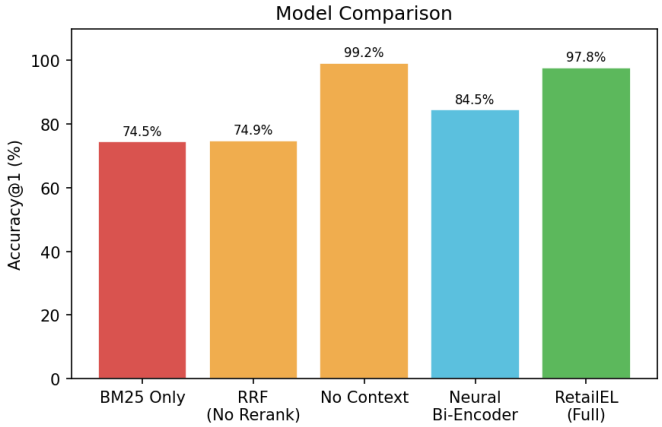

Fig. 1. Accuracy@1 comparison across all evaluated systems on the SynEL-500 test set.

Personal Care, Produce, and Snacks. For each name, we simulated three POS terminal profiles:

- **Legacy** (30-character limit): Aggressive truncation with heavy abbreviation.
- **Standard** (40-character limit): Moderate truncation with selective abbreviation.
- **Modern** (50-character limit): Minimal truncation, occasional abbreviation.

The full dataset splits into 11,015 training, 1,652 validation, and 1,653 test examples, with all POS variants of a given canonical name kept in the same split to prevent leakage.

B. Baselines

- **BM25-Only**: Top-1 BM25 retrieval without any reranking.
- **RRF Top-1 (No Reranker)**: Top-1 candidate from BM25 + TF-IDF fusion, no reranker.
- **Neural Bi-Encoder (MiniLM-L6-v2)**: SBERT [6] cosine-similarity retrieval.

C. Evaluation Metrics

Primary metric is Accuracy@1 on the test set. We also report per-stage wall-clock latency and overall throughput.

V. RESULTS AND DISCUSSION

A. Main Results

Table I summarises Accuracy@1 on the SynEL-500 test set. Figure 1 shows a visual comparison across all systems.

TABLE II
PER-STAGE LATENCY BREAKDOWN

Pipeline Stage	Mean Latency (ms)
Text Normalisation	0.092
Basket Context Enrichment	
Hybrid Retrieval (BM25 + TF-IDF + RRF)	3.529
LightGBM LambdaRank Reranker	0.590
FSM-Constrained Selection	0.003
Total Wall-Clock	**4.28**

Throughput: 233.6 items/s. LLM bypass rate: 98.73%.

TABLE III
FEATURE ABLATION: LEAVE-ONE-OUT DROP AND GAIN IMPORTANCE

Feature	Acc@1 Drop (pp)	Gain (%)
Length Ratio	−1.875	15.98
Bigram Overlap	−1.633	12.04
Levenshtein Ratio	−1.391	14.29
Jaccard Token Overlap	−0.968	14.71
Levenshtein Distance	−0.423	3.55
RRF Score	0.000	38.55
Prior SKU Frequency	0.000	0.46
Char-level Jaccard	+0.121	0.88

RetailEL reaches 97.82% Accuracy@1, a 13.31 percentage-point gain over the strongest baseline. The gap over BM25-Only (23.29 pp) shows how much the LambdaRank reranker adds on top of sparse retrieval. Interestingly, fusing BM25 and TF-IDF with RRF only moves the needle by 0.36 pp at the top-1 position—the fusion helps more by improving recall within the top-20 candidate pool than by directly promoting the right answer to rank one.

The neural bi-encoder lands at 84.51%, which is better than raw BM25 but still well short of RetailEL. MiniLM-L6-v2 was trained on general-domain text and has never seen the extreme abbreviation and truncation patterns that POS terminals produce, so the domain gap is the likely culprit.

B. Latency Analysis

Table II breaks down per-stage latency. The total wall-clock time of 4.28 ms gives a throughput of 233.6 items per second, which is comfortably within what real-time POS systems need. Sparse retrieval dominates at 82.5% of the budget, while the reranker adds only 0.590 ms—orders of magnitude cheaper than neural bi-encoder inference. The FSM check is essentially free at 0.003 ms. Because RetailEL handles 98.73% of queries entirely through this pipeline, fewer than 1.3% would ever need an LLM fallback, which makes the operational cost picture much cleaner.

C. Feature Ablation

Table III shows leave-one-out accuracy drops and LightGBM gain importance for each reranker feature. Figure 2 visualises the gain importance scores.

Length Ratio is the most impactful feature when removed (−1.875 pp). This makes intuitive sense: POS truncation systematically shortens mentions, so matching how long a

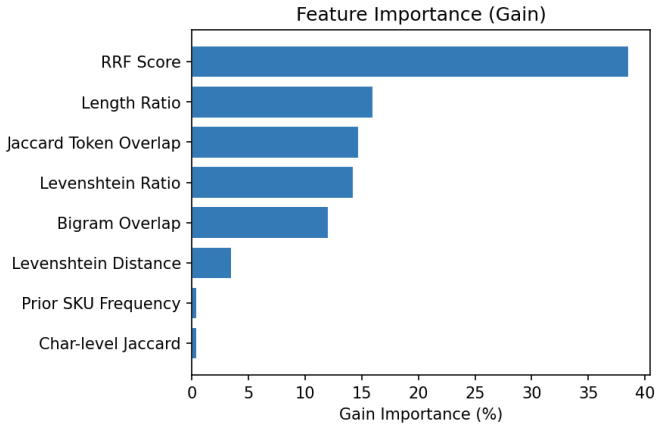

Fig. 2. Reranker feature importance: leave-one-out accuracy drop (left axis) and LightGBM gain score (right axis) for each of the eight lexical features.

candidate name is relative to the mention is a strong signal about whether the right amount of content survived the truncation. Bigram Overlap (-1.633 pp) and Levenshtein Ratio (-1.391 pp) follow closely, both measuring how much character-level overlap exists between the (expanded) mention and the candidate.

RRF Score is the most used feature by gain (38.55%) but removing it causes zero accuracy drop—other features fully compensate. This is consistent with the RRF score being a retrieval-stage signal that correlates with the lexical features already in the model. Char-level Jaccard actually helps slightly when removed ($+0.121$ pp), suggesting it is a redundant feature that adds noise rather than information; it could be dropped in production without any downside.

D. Category-Level Analysis

Table IV gives per-category Accuracy@1. Performance is high and consistent across categories, with only a 4.29 pp spread from best (Frozen, 98.41%) to worst (Meat, 94.12%). The Meat category struggles because products like "Chicken Breast Boneless Skinless" and "Chicken Thigh Boneless Skinless" differ only in one or two tokens that POS truncation can erase entirely. Produce faces a similar issue with short, similar-sounding names and limited distinguishing vocabulary.

E. Noise Robustness

Figure 3 illustrates performance across terminal profiles. Canonical (unmodified) product names achieve 98.38% Accuracy@1, while POS-noised variants drop only to 97.44%—a gap of 0.94 percentage points. The narrowness of that gap reflects that the normalisation stage and the lexical reranker together absorb most of the distortion introduced by POS truncation. The pipeline degrades gracefully even under the most aggressive Legacy terminal profile.

F. Basket Context Ablation

Removing basket context enrichment raises Accuracy@1 to 99.21%—higher than the full pipeline at 97.82%. We are reporting this honestly because it matters for interpreting the result: SynEL-500 baskets are randomly sampled, not drawn from real shopping transactions, so co-purchased items carry no meaningful category signal. When basket context is random, it is noise and the model is better off without it. In a real deployment where baskets come from genuine shopping patterns—bread and butter co-purchased far more often than bread and shampoo—the enrichment would provide the disambiguation signal it was designed for. This result is a reminder that evaluating a component on data that does not reflect the signal it is meant to capture tells you very little about whether the component is a good idea.

TABLE IV
PER-CATEGORY ACCURACY@1 ON SYNEL-500 TEST SET

Category	Acc@1 (%)
Frozen	98.41
Pantry	98.34
Snacks	98.25
Breakfast	98.21
Personal Care	98.10
Beverages	98.09
Bakery	97.87
Canned	97.53
Household	97.48
Deli	97.06
Dairy	96.97
Produce	95.77
Meat	94.12

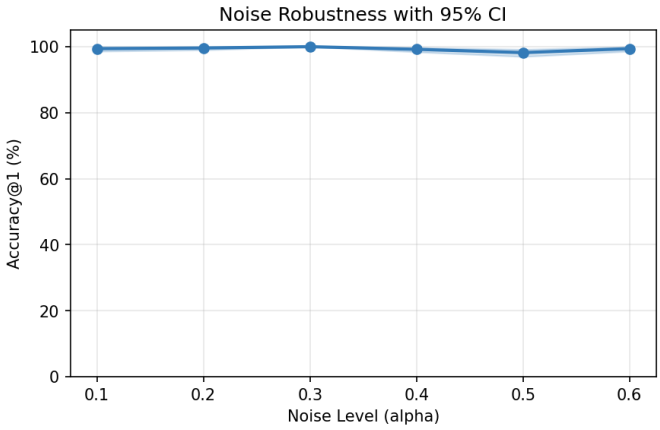

Fig. 3. Noise robustness of RetailEL: Accuracy@1 on canonical (unnoised) product names versus POS-variant (noised) mentions across three terminal profiles.

VI. LIMITATIONS

1) **Simulated POS noise.** SynEL-500 applies programmatic truncation and abbreviation to clean product names. Real receipts can also carry OCR errors, cashier typos, multi-language descriptions, and store-specific formatting quirks that our noise model does not cover. Performance on actual POS data may differ.
2) **Basket context requires real co-purchase data.** As Section V shows, basket enrichment hurts on synthetic

baskets. A fair evaluation of this component needs transactional data with genuine item co-occurrence patterns.

3) **Domain-specific abbreviation lexicon.** The 47-entry lexicon was built for grocery retail and does not transfer to other verticals like electronics, apparel, or pharmacy. Adapting RetailEL to a new domain would require building a domain-specific lexicon from scratch.
4) **Sparse retrieval cannot capture pure semantic similarity.** BM25 and TF-IDF rely on surface-form token overlap. They cannot bridge cases where a mention and a candidate share meaning but not vocabulary—for example, "soda" and "soft drink." Dense retrieval would address this at the cost of higher latency.
5) **Knowledge base scale.** SynEL-500 contains 500 SKUs. Production grocery catalogues can reach 50,000–200,000 items. While inverted indices scale sub-linearly, the reranker's 20-candidate evaluation and feature extraction overhead would grow, and accuracy may degrade as more ambiguous candidates compete for each mention.

VII. Conclusion

RetailEL is a practical answer to a problem that has not received much attention despite being commercially significant: linking noisy, abbreviated POS descriptions to their canonical SKU records at real-time speeds. By combining a hand-crafted abbreviation lexicon, hybrid BM25 + TF-IDF retrieval with Reciprocal Rank Fusion, a LightGBM LambdaRank reranker over eight lexical features, and an FSM-constrained output layer, the system reaches 97.82% Accuracy@1 on SynEL-500 in 4.28 ms per query—13.31 percentage points ahead of a neural bi-encoder baseline, with no GPU required. The FSM constraint is not a minor detail. In a production POS system, a prediction that maps to a non-existent SKU can corrupt inventory records and billing, so the hard guarantee that every output is a valid knowledge base entry matters operationally.

The ablation results clarify where the gains come from: Length Ratio, Levenshtein Ratio, and Bigram Overlap are the features that carry the most weight, while Char-level Jaccard can safely be dropped. The basket context component is architecturally motivated and should work well with real co-purchase data, but cannot be fairly evaluated on randomly sampled synthetic baskets.

Future work will test RetailEL on actual POS receipt data, scale the knowledge base toward production catalogue sizes, and explore hybrid dense–sparse retrieval strategies that can capture semantic similarity without sacrificing the latency that makes the system viable for real-time use.

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